RQ2: The Grid — Does Qualifying Predict the Race?

Research Question: *How strongly does grid (qualifying) position predict race finishing position, and does this differ between street circuits and permanent circuits?*

This notebook explores the relationship between starting position and finishing position across 25,000+ race entries. The scatter plot includes regression overlays, circuit-type filtering, and a Monaco 2024 spotlight.

Dataset: `rq2_grid_finish_circuits.csv` — individual race entries with grid position, finish position, circuit type, and race metadata.

In [1]:
```
import plotly.io as pio
pio.renderers.default = 'png'
pio.kaleido.scope.default_width = 900
pio.kaleido.scope.default_height = 520
import pandas as pd
import numpy as np
import plotly.express as px
import plotly.graph_objects as go
from plotly.subplots import make_subplots
from scipy import stats

pd.set_option('display.max_rows', 100)

df = pd.read_csv('../final_datasets/rq2_grid_finish_circuits.csv')
print(f"Shape: {df.shape}")
print(f"\nColumn types:\n{df.dtypes}")
print(f"\nFirst 5 rows:")
df.head()
```

```
/var/folders/9r/jpz6t9fx7jv6tyyxhcmg5k200000gn/T/ipykernel_67789/2972423914.
py:3: DeprecationWarning:
Use of plotly.io.kaleido.scope.default_width is deprecated and support will
be removed after September 2025.
Please use plotly.io.defaults.default_width instead.

  pio.kaleido.scope.default_width = 900
/var/folders/9r/jpz6t9fx7jv6tyyxhcmg5k200000gn/T/ipykernel_67789/2972423914.
py:4: DeprecationWarning:
Use of plotly.io.kaleido.scope.default_height is deprecated and support will
be removed after September 2025.
Please use plotly.io.defaults.default_height instead.

  pio.kaleido.scope.default_height = 520
```

```
Shape: (25238, 10)

Column types:
Year                 int64
Race Name              str
Circuit Name           str
Driver                 str
Constructor            str
Grid Position      float64
Finish Position    float64
Points             float64
Dataset                str
Circuit Type           str
dtype: object

First 5 rows:
```

Out[1]:

	Year	Race Name	Circuit Name	Driver	Constructor	Grid Position	Finish Position	Points	Datas
0	2008	Australian GP	Albert Park Grand Prix Circuit	Lewis Hamilton	McLaren	1.0	1.0	10.0	Histori (195 20
1	2008	Australian GP	Albert Park Grand Prix Circuit	Nick Heidfeld	Sauber	5.0	2.0	8.0	Histori (195 20
2	2008	Australian GP	Albert Park Grand Prix Circuit	Nico Rosberg	Williams	7.0	3.0	6.0	Histori (195 20
3	2008	Australian GP	Albert Park Grand Prix Circuit	Fernando Alonso	Renault	11.0	4.0	5.0	Histori (195 20
4	2008	Australian GP	Albert Park Grand Prix Circuit	Heikki Kovalainen	McLaren	3.0	5.0	4.0	Histori (195 20

1. Data Quality Check

In [2]:

```python
print("Missing values per column:")
print(df.isnull().sum())
print(f"\nDuplicate rows: {df.duplicated().sum()}")
print(f"\nYear range: {df['Year'].min()} – {df['Year'].max()}")
print(f"Unique drivers: {df['Driver'].nunique()}")
```

```
print(f"Unique tracks (Race Name): {df['Race Name'].nunique()}")
print(f"\nCircuit Type distribution:")
print(df['Circuit Type'].value_counts())
```

```
Missing values per column:
Year                0
Race Name           0
Circuit Name        0
Driver              0
Constructor         0
Grid Position       0
Finish Position     0
Points              0
Dataset             0
Circuit Type        0
dtype: int64

Duplicate rows: 0

Year range: 1950 – 2026
Unique drivers: 799
Unique tracks (Race Name): 46

Circuit Type distribution:
Circuit Type
Permanent Circuit    22289
Street Circuit        2949
Name: count, dtype: int64
```

2. Descriptive Statistics

In [3]:
```
print("Descriptive statistics:\n")
df[['Grid Position', 'Finish Position', 'Points']].describe().round(2)
```

```
Descriptive statistics:
```

Out[3]:

	Grid Position	Finish Position	Points
count	25238.00	25238.00	25238.00
mean	11.86	11.72	2.21
std	6.85	6.80	4.57
min	1.00	1.00	0.00
25%	6.00	6.00	0.00
50%	12.00	11.00	0.00
75%	17.00	17.00	2.00
max	71.00	34.00	50.00

3. Position Change Analysis

Creating a derived column: `Position Change = Finish Position - Grid Position`. Negative means gained positions, positive means lost positions.

In [4]:
```
# Filter valid entries (same as D3 does)
valid = df[(df['Grid Position'] > 0) & (df['Finish Position'] > 0) &
           (df['Grid Position'] <= 26) & (df['Finish Position'] <= 26)].copy

valid['Position Change'] = valid['Finish Position'] - valid['Grid Position']

print(f"Valid entries: {len(valid)} out of {len(df)}")
print(f"\nCorrelation (Grid vs Finish):")
print(f"  Overall:   r = {valid['Grid Position'].corr(valid['Finish Positior
print(f"  Street:    r = {valid[valid['Circuit Type'] == 'Street Circuit']['
print(f"  Permanent: r = {valid[valid['Circuit Type'] == 'Permanent Circuit'

print(f"\nPosition change stats:")
valid['Position Change'].describe().round(2)
```

```
Valid entries: 24956 out of 25238

Correlation (Grid vs Finish):
  Overall:   r = 0.4524
  Street:    r = 0.4371
  Permanent: r = 0.4542

Position change stats:
```

Out[4]:
```
count    24956.00
mean        -0.13
std          7.00
min        -23.00
25%         -4.00
50%         -1.00
75%          3.00
max         25.00
Name: Position Change, dtype: float64
```

Visualization Prototyping

4. Grid and Finish Position Distributions

In [5]:
```
fig = make_subplots(rows=1, cols=2, subplot_titles=('Grid Position Distribut

fig.add_trace(go.Histogram(
    x=valid['Grid Position'], nbinsx=26,
    marker_color='#4ecdc4', marker_line_color='#0a0a0a', marker_line_width=0
    name='Grid'
), row=1, col=1)

fig.add_trace(go.Histogram(
    x=valid['Finish Position'], nbinsx=26,
```

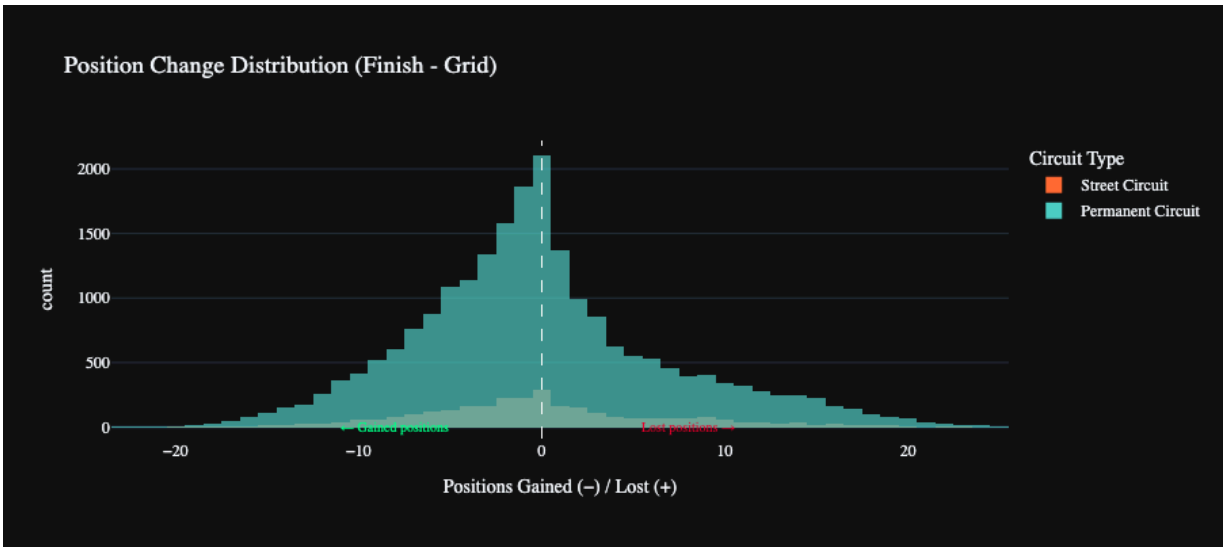
```
    marker_color='#ff6b35', marker_line_color='#0a0a0a', marker_line_width=0
    name='Finish'
), row=1, col=2)

fig.update_layout(
    title='Distribution of Grid vs Finish Positions',
    template='plotly_dark', height=380,
    font=dict(family='DM Sans'), title_font_size=18,
    showlegend=False
)
fig.show()
```

5. Position Change Distribution

How many positions do drivers typically gain or lose during a race? Negative = gained positions, positive = dropped.

In [6]:
```
fig = px.histogram(
    valid, x='Position Change', nbins=50,
    color='Circuit Type',
    color_discrete_map={'Street Circuit': '#ff6b35', 'Permanent Circuit': '#
    barmode='overlay',
    title='Position Change Distribution (Finish - Grid)',
    labels={'Position Change': 'Positions Gained (-) / Lost (+)'},
    template='plotly_dark', opacity=0.7
)
fig.add_vline(x=0, line_dash='dash', line_color='white', line_width=1)
fig.update_layout(
    font=dict(family='DM Sans'), title_font_size=18, height=400,
    annotations=[dict(
        x=-8, y=0, xref='x', yref='y',
        text='← Gained positions', showarrow=False,
        font=dict(size=10, color='#00ff87')
    ), dict(
        x=8, y=0, xref='x', yref='y',
        text='Lost positions →', showarrow=False,
        font=dict(size=10, color='#e8002d')
    )]
```

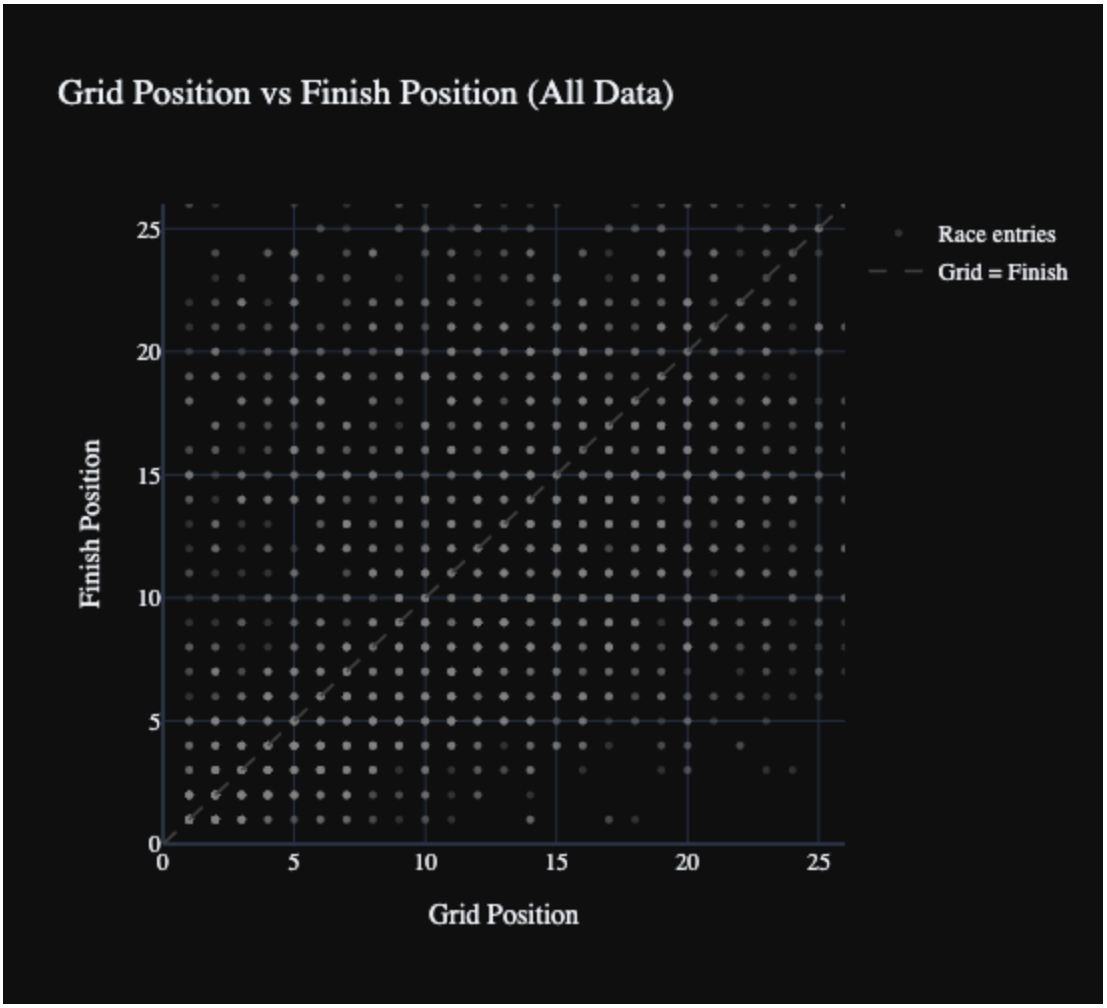
```
)
fig.show()
```



6. Basic Scatter — Grid vs Finish (All Data)

The raw prototype — every race entry plotted. The diagonal dashed line represents "grid = finish" (no position change). Points above the line lost positions; below gained.

In [7]:
```
# Sample for performance (same approach as D3)
sample = valid.sample(n=min(3000, len(valid)), random_state=42)

fig = go.Figure()

fig.add_trace(go.Scatter(
    x=sample['Grid Position'], y=sample['Finish Position'],
    mode='markers',
    marker=dict(size=4, color='#888888', opacity=0.3),
    name='Race entries'
))

# Diagonal reference
fig.add_trace(go.Scatter(
    x=[0, 26], y=[0, 26],
    mode='lines',
    line=dict(dash='dash', color='#444444', width=1.5),
    name='Grid = Finish',
    showlegend=True
))

fig.update_layout(
    title='Grid Position vs Finish Position (All Data)',
    xaxis_title='Grid Position', yaxis_title='Finish Position',
    xaxis=dict(range=[0, 26]), yaxis=dict(range=[0, 26]),
    template='plotly_dark', height=500, width=550,
    font=dict(family='DM Sans'), title_font_size=18,
)
fig.show()
```

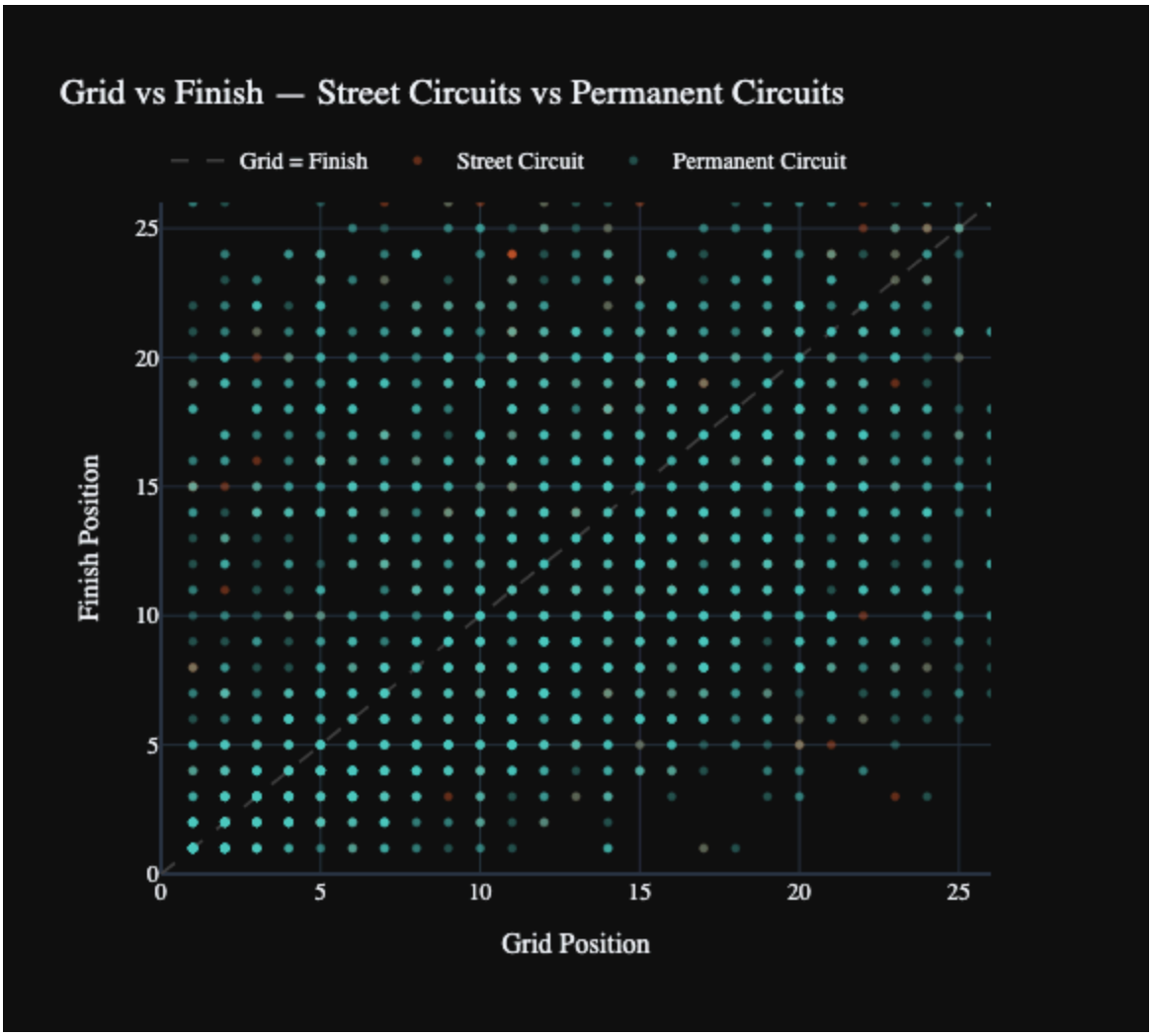


7. Scatter Colored by Circuit Type

Splitting by circuit type reveals the core insight: street circuits (orange) cluster tighter around the diagonal than permanent circuits (teal).

In [8]:
```
STREET_COLOR = '#ff6b35'
PERMANENT_COLOR = '#4ecdc4'

fig = go.Figure()

# Diagonal reference
fig.add_trace(go.Scatter(
    x=[0, 26], y=[0, 26], mode='lines',
    line=dict(dash='dash', color='#444444', width=1.5),
    name='Grid = Finish'
))

for ct, color in [('Street Circuit', STREET_COLOR), ('Permanent Circuit', PE
    subset = sample[sample['Circuit Type'] == ct]
    fig.add_trace(go.Scatter(
        x=subset['Grid Position'], y=subset['Finish Position'],
        mode='markers', name=ct,
        marker=dict(size=4, color=color, opacity=0.35,
```

```
                        line=dict(color=color, width=0.5)),
    ))

fig.update_layout(
    title='Grid vs Finish — Street Circuits vs Permanent Circuits',
    xaxis_title='Grid Position', yaxis_title='Finish Position',
    xaxis=dict(range=[0, 26]), yaxis=dict(range=[0, 26]),
    template='plotly_dark', height=520, width=580,
    font=dict(family='DM Sans'), title_font_size=18,
    legend=dict(orientation='h', yanchor='bottom', y=1.02)
)
fig.show()
```

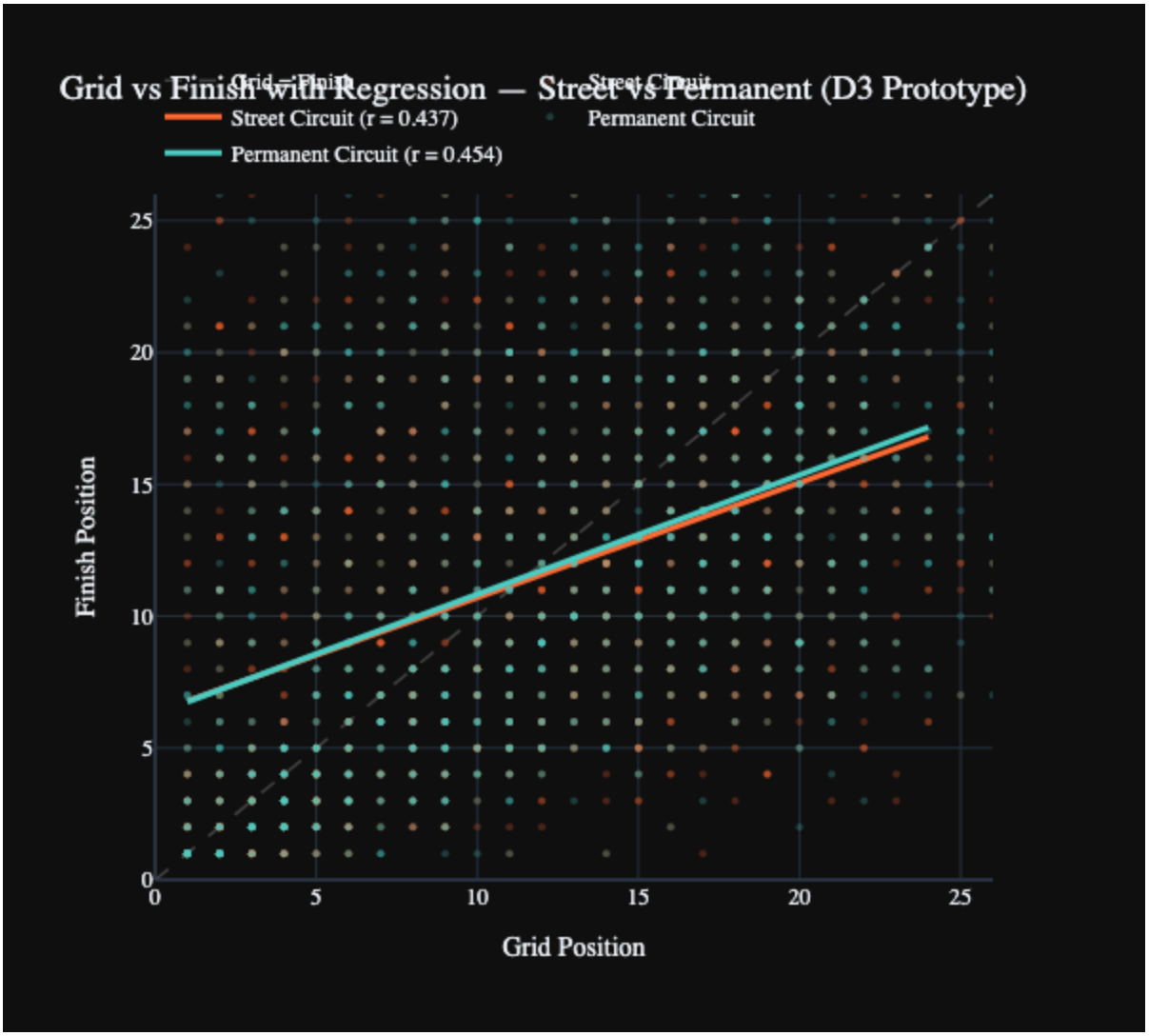


8. Scatter with Regression Lines and R-values

Adding linear regression per circuit type — The regression slope and Pearson r show how predictive qualifying is for each circuit type.

In [9]:
```
fig = go.Figure()

# Diagonal
fig.add_trace(go.Scatter(
    x=[0, 26], y=[0, 26], mode='lines',
```

```
    line=dict(dash='dash', color='#444444', width=1.5),
    name='Grid = Finish', showlegend=True
))

for ct, color in [('Street Circuit', STREET_COLOR), ('Permanent Circuit', PE
    subset = valid[valid['Circuit Type'] == ct]
    sample_ct = subset.sample(n=min(1500, len(subset)), random_state=42)

    # Scatter
    fig.add_trace(go.Scatter(
        x=sample_ct['Grid Position'], y=sample_ct['Finish Position'],
        mode='markers', name=ct,
        marker=dict(size=4, color=color, opacity=0.25),
    ))

    # Regression line
    slope, intercept, r_value, p_value, std_err = stats.linregress(
        subset['Grid Position'], subset['Finish Position']
    )
    x_reg = np.array([1, 24])
    y_reg = slope * x_reg + intercept

    fig.add_trace(go.Scatter(
        x=x_reg, y=y_reg, mode='lines',
        line=dict(color=color, width=3),
        name=f'{ct} (r = {r_value:.3f})',
    ))

fig.update_layout(
    title='Grid vs Finish with Regression — Street vs Permanent (D3 Prototyp
    xaxis_title='Grid Position', yaxis_title='Finish Position',
    xaxis=dict(range=[0, 26]), yaxis=dict(range=[0, 26]),
    template='plotly_dark', height=540, width=600,
    font=dict(family='DM Sans'), title_font_size=18,
    legend=dict(orientation='h', yanchor='bottom', y=1.02)
)
fig.show()
```

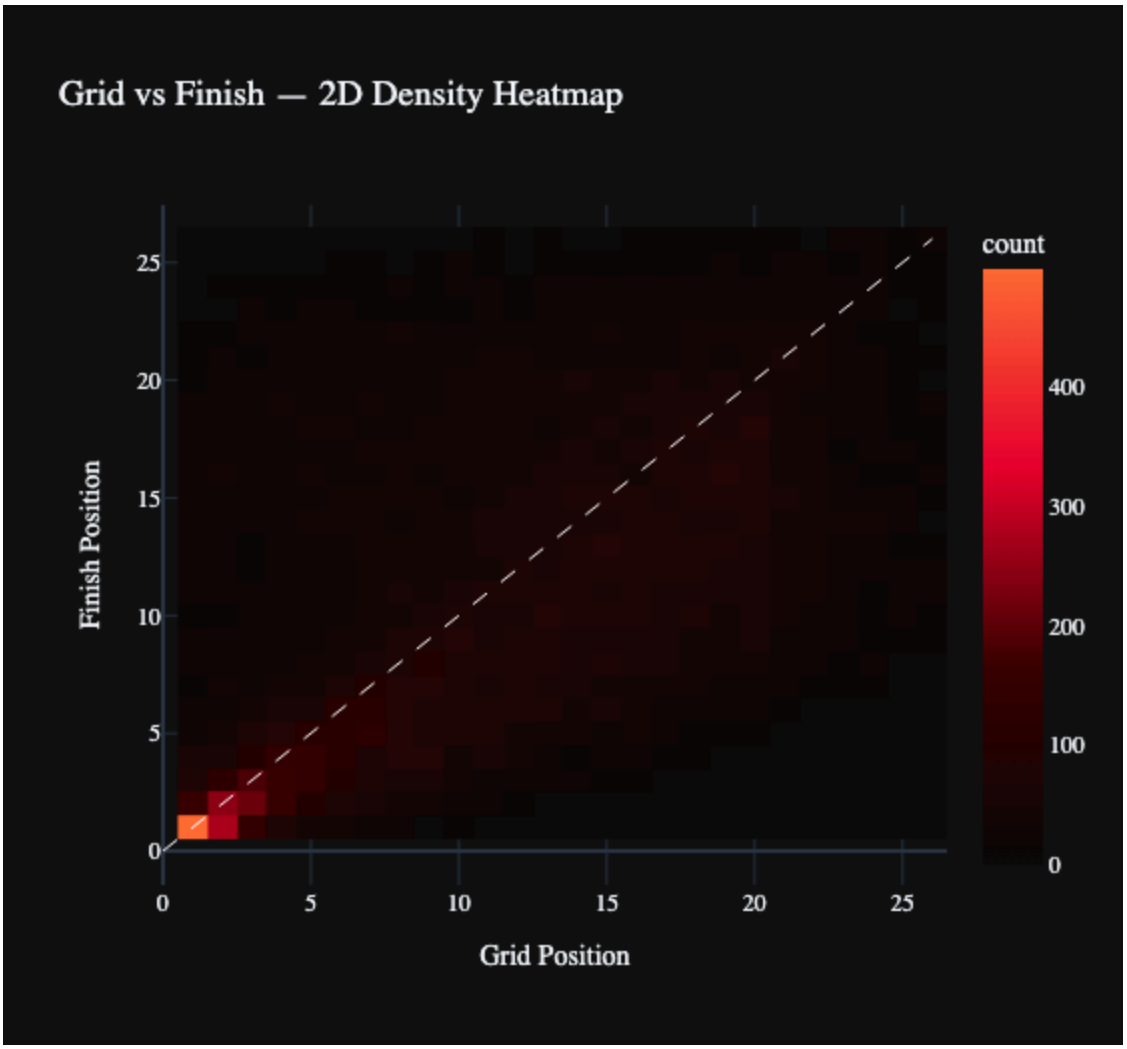


9. 2D Density Heatmap — Grid vs Finish Frequency

With 25,000+ data points, scatter dots overlap heavily. A 2D histogram / heatmap shows the true density — which grid→finish combinations are most common.

In [10]:
```
fig = px.density_heatmap(
    valid, x='Grid Position', y='Finish Position',
    nbinsx=26, nbinsy=26,
    color_continuous_scale=['#0a0a0a', '#3a0000', '#e8002d', '#ff6b35'],
    title='Grid vs Finish — 2D Density Heatmap',
    template='plotly_dark'
)

# Add diagonal reference
fig.add_trace(go.Scatter(
    x=list(range(0, 27)), y=list(range(0, 27)),
    mode='lines', line=dict(dash='dash', color='white', width=1),
    showlegend=False
))

fig.update_layout(
```

```
    font=dict(family='DM Sans'), title_font_size=18, height=520, width=560,
    xaxis_title='Grid Position', yaxis_title='Finish Position',
)
fig.show()
```

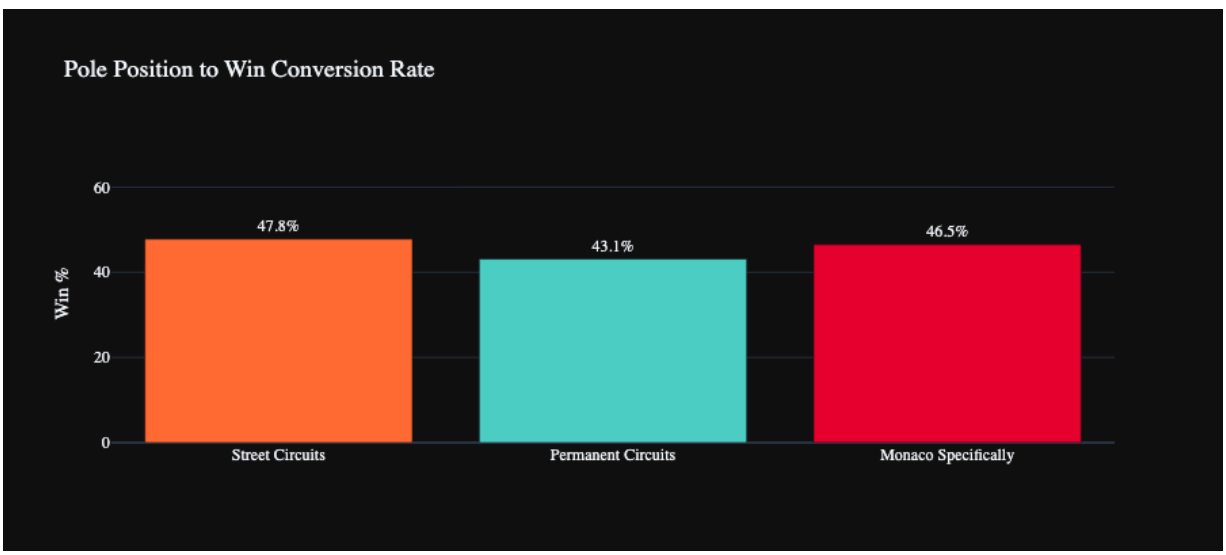


10. Pole-to-Win Conversion Rates

Key stats: what percentage of pole sitters (P1 on the grid) convert that into a win, broken down by circuit type.

In [11]:
```
pole_sitters = valid[valid['Grid Position'] == 1]

street_pole = pole_sitters[pole_sitters['Circuit Type'] == 'Street Circuit']
perm_pole = pole_sitters[pole_sitters['Circuit Type'] == 'Permanent Circuit'
monaco_pole = pole_sitters[pole_sitters['Race Name'].str.contains('Monaco',

street_win = (street_pole['Finish Position'] == 1).sum() / len(street_pole)
perm_win = (perm_pole['Finish Position'] == 1).sum() / len(perm_pole) * 100
monaco_win = (monaco_pole['Finish Position'] == 1).sum() / len(monaco_pole)

stats_data = pd.DataFrame({
    'Category': ['Street Circuits', 'Permanent Circuits', 'Monaco Specifical
    'Pole-to-Win %': [round(street_win, 1), round(perm_win, 1), round(monacc
```

```
    'Total Poles': [len(street_pole), len(perm_pole), len(monaco_pole)]
})

fig = go.Figure(go.Bar(
    x=stats_data['Category'],
    y=stats_data['Pole-to-Win %'],
    marker_color=[STREET_COLOR, PERMANENT_COLOR, '#e8002d'],
    text=stats_data['Pole-to-Win %'].astype(str) + '%',
    textposition='outside'
))
fig.update_layout(
    title='Pole Position to Win Conversion Rate',
    yaxis_title='Win %', yaxis=dict(range=[0, 70]),
    template='plotly_dark', height=400,
    font=dict(family='DM Sans'), title_font_size=18,
)
fig.show()

print("\nDetailed stats:")
stats_data
```




```
Detailed stats:
```

Out[11]:

	Category	Pole-to-Win %	Total Poles
0	Street Circuits	47.8	136
1	Permanent Circuits	43.1	1006
2	Monaco Specifically	46.5	71

11. Monaco 2024 Spotlight

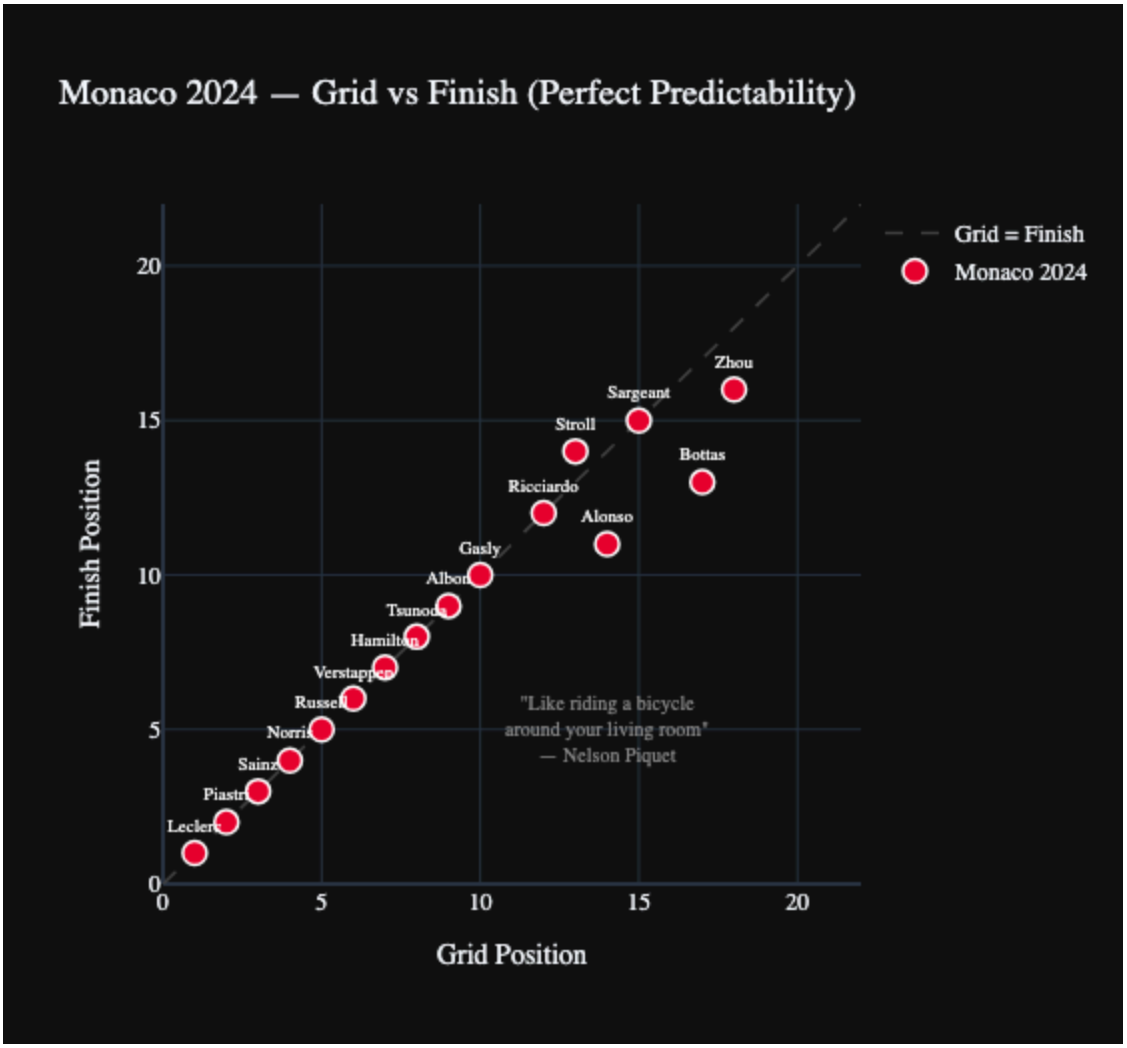
The most extreme example of grid predictability in recent history — at Monaco 2024, the top 10 qualifiers finished in the exact same order. Zero overtakes in the top 10.

In [12]:

```
monaco24 = valid[(valid['Year'] == 2024) & (valid['Race Name'].str.contains(
print(f"Monaco 2024 entries: {len(monaco24)}")
print(f"Entries where Grid == Finish: {(monaco24['Grid Position'] == monaco2
```

```
print()

fig = go.Figure()

# Diagonal
fig.add_trace(go.Scatter(
    x=[0, 22], y=[0, 22], mode='lines',
    line=dict(dash='dash', color='#444444', width=1.5),
    name='Grid = Finish'
))

# Monaco 2024 dots
fig.add_trace(go.Scatter(
    x=monaco24['Grid Position'], y=monaco24['Finish Position'],
    mode='markers+text',
    marker=dict(size=12, color='#e8002d', symbol='circle',
                line=dict(color='white', width=1.5)),
    text=monaco24['Driver'].apply(lambda x: x.split()[-1]),
    textposition='top center',
    textfont=dict(size=9, color='white'),
    name='Monaco 2024'
))

fig.update_layout(
    title='Monaco 2024 — Grid vs Finish (Perfect Predictability)',
    xaxis_title='Grid Position', yaxis_title='Finish Position',
    xaxis=dict(range=[0, 22]), yaxis=dict(range=[0, 22]),
    template='plotly_dark', height=520, width=560,
    font=dict(family='DM Sans'), title_font_size=18,
    annotations=[dict(
        x=14, y=5, text='"Like riding a bicycle<br>around your living room"<
        showarrow=False, font=dict(size=10, color='#888888', family='DM Sans
    )]
)
fig.show()

monaco24[['Driver', 'Grid Position', 'Finish Position', 'Position Change']].
```

```
Monaco 2024 entries: 16
Entries where Grid == Finish: 12
```


Monaco 2024 — Grid vs Finish (Perfect Predictability)
Grid = Finish
Monaco 2024
Finish Position
Grid Position
0
5
10
15
20
Zhou
Sargeant
Stroll
Bottas
Ricciardo
Alonso
Gasly
Albon
Tsunoda
Hamilton
Verstappen
Russell
Norris
Sainz
Piastri
Leclerc
"Like riding a bicycle
around your living room"
— Nelson Piquet

Out[12]:

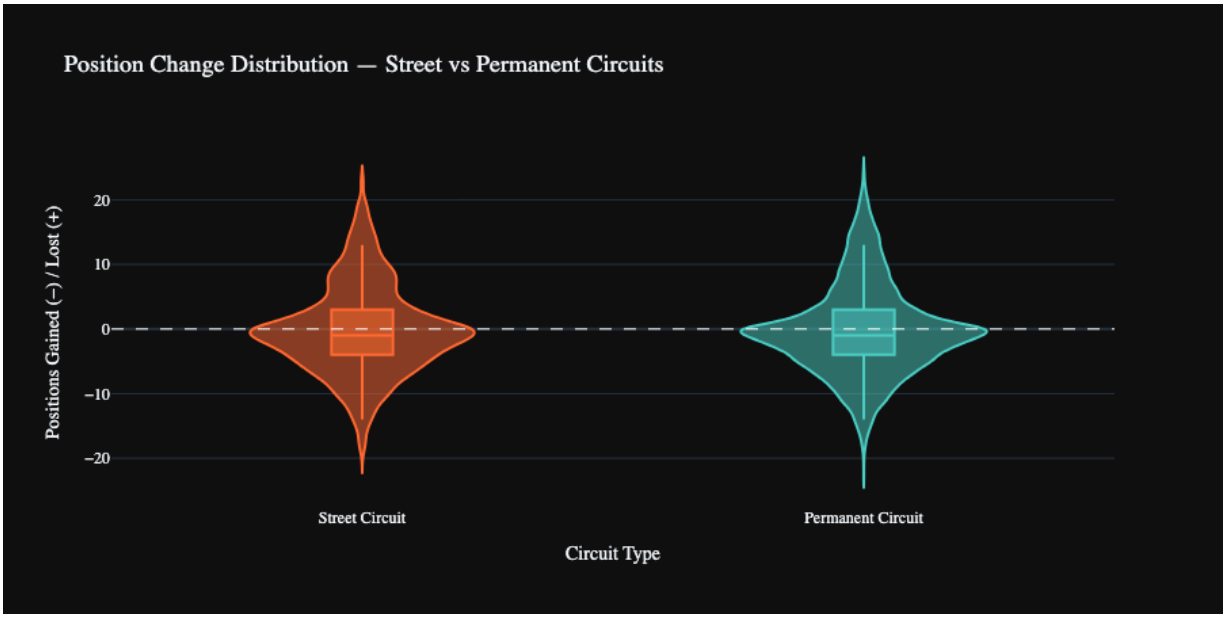
	Driver	Grid Position	Finish Position	Position Change
24494	Charles Leclerc	1.0	1.0	0.0
24495	Oscar Piastri	2.0	2.0	0.0
24496	Carlos Sainz	3.0	3.0	0.0
24497	Lando Norris	4.0	4.0	0.0
24498	George Russell	5.0	5.0	0.0
24499	Max Verstappen	6.0	6.0	0.0
24500	Lewis Hamilton	7.0	7.0	0.0
24501	Yuki Tsunoda	8.0	8.0	0.0
24502	Alexander Albon	9.0	9.0	0.0
24503	Pierre Gasly	10.0	10.0	0.0
24505	Daniel Ricciardo	12.0	12.0	0.0
24507	Lance Stroll	13.0	14.0	1.0
24504	Fernando Alonso	14.0	11.0	-3.0
24508	Logan Sargeant	15.0	15.0	0.0
24506	Valtteri Bottas	17.0	13.0	-4.0
24509	Guanyu Zhou	18.0	16.0	-2.0

12. Violin Plot — Position Changes by Circuit Type

Comparing the spread of position changes between street and permanent circuits. Street circuits should show a tighter distribution (less overtaking opportunity).

In [13]:

```
fig = px.violin(
    valid, x='Circuit Type', y='Position Change',
    color='Circuit Type',
    color_discrete_map={'Street Circuit': STREET_COLOR, 'Permanent Circuit':
    box=True, points=False,
    title='Position Change Distribution — Street vs Permanent Circuits',
    template='plotly_dark'
)
fig.add_hline(y=0, line_dash='dash', line_color='white', line_width=1)
fig.update_layout(
    font=dict(family='DM Sans'), title_font_size=18, height=450,
    showlegend=False,
    yaxis_title='Positions Gained (−) / Lost (+)'
)
fig.show()
```

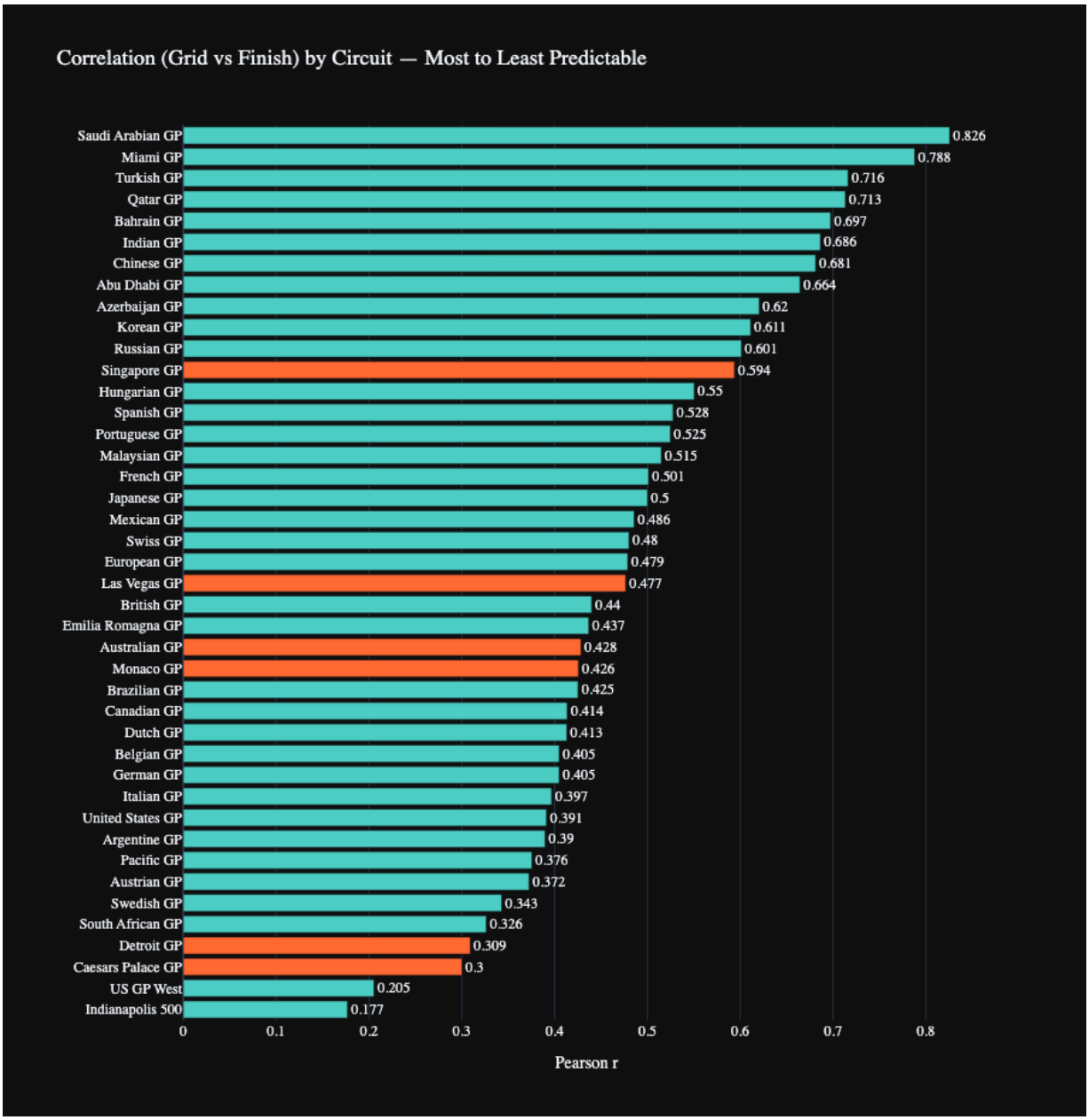


13. Correlation by Track — Most vs Least Predictable Circuits

Calculating the Pearson correlation (R) between grid and finish position for each individual track. Tracks with higher R are more predictable (qualifying determines the race).

In [14]:
```
# Min 50 entries per track
track_corr = valid.groupby('Race Name').apply(
    lambda g: pd.Series({
        'r': g['Grid Position'].corr(g['Finish Position']),
        'entries': len(g),
        'circuit_type': g['Circuit Type'].mode().iloc[0] if len(g) > 0 else
    })
).reset_index()
track_corr = track_corr[track_corr['entries'] >= 50].sort_values('r', ascenc

colors = [STREET_COLOR if ct == 'Street Circuit' else PERMANENT_COLOR
          for ct in track_corr['circuit_type']]

fig = go.Figure(go.Bar(
    y=track_corr['Race Name'],
    x=track_corr['r'],
    orientation='h',
    marker_color=colors,
    text=track_corr['r'].round(3), textposition='outside'
))
fig.update_layout(
    title='Correlation (Grid vs Finish) by Circuit — Most to Least Predictab
    xaxis_title='Pearson r', yaxis_title='',
    template='plotly_dark', height=max(400, len(track_corr) * 22),
    font=dict(family='DM Sans'), title_font_size=18,
    margin=dict(l=150)
```

```
)
fig.show()
```



Key Findings

1. **Grid position is a strong predictor of finish position** — the overall correlation is significant across all circuit types.
2. **Street circuits are more predictable** than permanent circuits — narrower streets make overtaking nearly impossible.
3. **Monaco is the extreme** — the pole-to-win conversion rate at Monaco exceeds other street circuits, and Monaco 2024 demonstrated this perfectly with zero top-10 position changes.

4. **The 2D density heatmap** reveals that the most common outcome is finishing within 2-3 positions of your grid slot, with the heaviest density along the diagonal.
5. **Per-track correlation** shows meaningful variance — some circuits (likely Monaco, Singapore) consistently show higher predictability than others.

Key design choices: circuit type color coding (orange/teal), regression line overlays with r-values, the Monaco 2024 highlight, and the stat cards showing pole-to-win conversion rates.

In [ ]: